

VARIANCE OF DISTRIBUTION OF PRIMES IN RESIDUE CLASSES

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1. Introduction

THE prime number theorem for arithmetic progressions tells us that, for integers $a, q \geq 1$ with $(a, q) = 1$ we have

$$E(x; q, a) = o\left(\frac{x}{\phi(q)}\right) \quad (1.1)$$

as x approaches infinity, where E is the error term

$$E(x; q, a) = \psi(x; q, a) - \frac{x}{\phi(q)} \quad (1.2)$$

in the ‘prime’ counting function

$$\psi(x; q, a) = \sum_{\substack{n \leq x \\ n \equiv a(q)}} \Lambda(n)$$

and where we must impose some restriction on the size of q that it not grow too quickly as a function of x . The best that can be said so far is that, for any fixed A , (1.1) holds subject to $q < (\log x)^A$, the Siegel–Walfisz theorem. Assuming the Generalized Riemann Hypothesis (GRH), one may relax this to $q < x^{\frac{1}{2}}(\log x)^{-2-\epsilon}$ and in fact the error is then bounded by

$$E(x; q, a) \ll x^{\frac{1}{2}}(\log x)^2. \quad (1.3)$$

Both for its own interest and for the sake of applications one would like to have information about progressions to larger moduli, say q right up to x (of course for q larger than x there is at most one integer in the progression). With this in mind one may consider averages for the error term $E(x; q, a)$ with respect to either or both of a and q . (We shall not mention the problem of averaging with respect to x although this too is not without interest.) In the case that one averages over q one obtains

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theorems of Bombieri–Vinogradov type which roughly speaking give the asymptotic formula on average over q with q as large as $x^{\frac{1}{2}}$, or with somewhat larger exponents for certain weighted versions, occurring in applications. (See as a reference § 12 of [2].)

In this paper we consider the problem for given q of averaging over a . We study the mean square sum

$$G(x, q) = \sum_{a(q)}^* E^2(x; q, a) \quad (1.4)$$

where the asterisk denotes the restriction of the summation to reduced residue classes.

The first known result on $G(x, q)$, due to Turan [20], is an upper bound (see also Montgomery [17]), namely: Under the assumption of GRH, we have

$$G(x, q) \ll x(\log x)^4. \quad (1.5)$$

The problem of obtaining an unconditional upper bound even remotely approaching the quality of (1.5) seems likely to prove quite difficult, at least for certain ranges of q . In the event, for example, that q is about $\exp(\sqrt{\log x})$ in size (actually a more general statement can be shown), and q is the modulus of a Dirichlet character whose L -function possesses an “exceptional” zero, then it follows easily from the explicit formula for $\psi(x; q, a)$, cf [4, chap. 20], that $G(x, q) \gg \frac{x^2}{\phi(q)} x^{-2(1-\beta)}$ which would be quite close to the trivial upper bound $\frac{x^2}{\phi(q)}$.

Far less is known about the behaviour of individual $G(x, q)$ than is known about the sum we get by averaging also over q to obtain

$$H(x, Q) = \sum_{q \leq Q} G(x, q). \quad (1.6)$$

The sum H has a long history [10] and results concerning it are usually described as the “Barban–Davenport–Halberstam Theorem” in honour of the works [1], [5]. The version of this pertinent to us is the Barban–Montgomery asymptotic formula [16]: For $Q \leq x$, any $A > 0$, any $\epsilon > 0$,

$$H(x, Q) = Qx \log Q + O(Qx) + O(x^2(\log x)^{-A}), \quad (1.7)$$

and the theorem of Hooley [11] that, on GRH,

$$H(x, Q) = Qx \log Q - cQx + O(Q^{\frac{1}{2}}x^{\frac{3}{2}}) + O(x^{\frac{1}{2}+\epsilon}), \quad (1.8)$$

for a suitable constant c ; see Theorem 4 below. These results suggest the

conjecture, made by Hooley [13, p. 362], that, subject to certain (unspecified) conditions,

$$G(x, q) \sim x \log q \quad (HC)$$

for some range of q . In this direction, Hooley [12] adapted Montgomery's method to prove (HC), for almost all q , $\frac{Q}{2} < q \leq Q$ in the range

$$x(\log x)^{-A} < Q \leq x \quad (1.9)$$

and, on GRH, the same result in the range

$$x^{\frac{1}{3}+\epsilon} \leq Q \leq x. \quad (1.10)$$

Our first result gives a sharper statement, and in the GRH case, in a wider range.

THEOREM 1. *Assume the Generalized Riemann Hypothesis. Then for $x^{\frac{1}{2}} \leq Q \leq x$, we have*

$$\sum_{\substack{q < Q \\ q \leq Q}} \left| G(x, q) - x \log q + x \left(\gamma + \log 2\pi + \sum_{p|q} \frac{\log p}{p-1} \right) \right| \\ \ll x^{\frac{1}{2}} (\log x)^{\frac{1}{2}} + \min \{ Q^{\frac{1}{2}} x^{\frac{1}{2}} (\log x)^{\frac{1}{2}}, Qx \}. \quad (1.11)$$

This holds unconditionally if the first term on the right of (1.11) is replaced by $x^2 (\log x)^{-A}$.

COROLLARY. *Assume the Generalized Riemann Hypothesis. Let $\epsilon > 0$, $x^{\frac{1}{2}+\epsilon} \leq Q \leq x$. Then, for almost all q with $\frac{Q}{2} \leq q \leq Q$, we have $G(x, q) \sim x \log q$.*

Theorem 1 suggests a refinement of (HC). We make the following:

CONJECTURE 1. *Let $\epsilon > 0$. For $x^{\frac{1}{2}+\epsilon} \leq q \leq x^{1-\epsilon}$,*

$$G(x, q) = x \log q - x \left(\gamma + \log 2\pi + \sum_{p|q} \frac{\log p}{p-1} \right) + O(x^{1-\delta(\epsilon)}). \quad (1.12)$$

We also believe:

CONJECTURE 2. *The asymptotic formula (HC) holds in the wider range $x^{\frac{1}{2}+\epsilon} \leq q \leq x$.*

It may well be that these also hold for smaller q , but below $q = x^{\frac{1}{2}}$ we are somewhat skeptical.

Our main interest here is the provision of further evidence for these conjectures. Although the above results are suggestive they do not, except for (1.5) deal with individual moduli other than in a statistical sense. Our second theorem is a lower bound which complements the upper bound (1.5) but is sharper, giving the right order of magnitude for all q in the same ranges as in Theorem 1.

If we let $q = x^\alpha$, $0 \leq \alpha \leq 1$, then the conjecture (HC) takes the form

$$G(x, q) \sim x \log q \sim \alpha x \log x.$$

We prove the following theorem

THEOREM 2. *Let $A > 0$, $\epsilon > 0$. We have, for x sufficiently large,*

$$G(x, q) \geq \left(\frac{1}{2} - \epsilon\right) x \log q \quad \text{for } x(\log x)^{-A} \leq q \leq x, \quad (1.13)$$

and, assuming the Generalized Riemann Hypothesis,

$$G(x, q) \geq 2\left(\alpha - \frac{3}{4} - \epsilon\right) x \log x = \left(2 - \frac{3}{2\alpha} - \epsilon\right) x \log q \quad \text{for } x^{1+\epsilon} \leq q \leq x. \quad (1.14)$$

If one assumes the Riemann Hypothesis and also a strong version of the twin prime conjecture then one can conclude that Conjectures 1 and 2 hold. Let

$$N_1 = N_1(k) = \max(0, -k), \quad N_2 = N_2(x, k) = \min(x, x - k), \quad (1.15)$$

and

$$E(x, k) = \sum_{N_1(k) < n \leq N_2(x, k)} \Lambda(n) \Lambda(n + k) - \mathfrak{S}(k)(x - |k|), \quad (1.16)$$

where we define as usual

$$\mathfrak{S}(k) = \begin{cases} 2C \prod_{\substack{p|k \\ p>2}} \left(\frac{p-1}{p-2}\right), & \text{if } k \text{ is even, } k \neq 0; \\ 0, & \text{if } k \text{ is odd;} \end{cases}$$

with

$$C = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right).$$

THEOREM 3. *Assume the Riemann Hypothesis, and assume for $0 < |k| \leq x$ and $\epsilon > 0$ that*

$$\sum_{N_1(k) < n \leq N_2(x, k)} \Lambda(n) \Lambda(n + k) = \mathfrak{S}(k)(x - |k|) + O(x^{1+\epsilon}). \quad (1.17)$$

Then Conjectures 1 and 2 hold as stated.

The starting point for our investigation is the following result.

PROPOSITION 1. *We have, for $q \geq 1$, with $R(x) = \psi(x) - x$,*

$$G(x, q) = x \log x - x + \sum_{0 < |j| \leq \frac{x}{q}} \sum_{N_1(jq) < n \leq N_2(x, jq)} \Lambda(n) \Lambda(n + jq) - \frac{x^2}{\phi(q)} - \frac{2x}{\phi(q)} R(x) + E \quad (1.18)$$

where

$$E \ll \frac{x(\log x)^2}{\phi(q)} + \left(x^{\frac{1}{2}} + \max_{u \leq x} |R(u)| \right) \log x + (\log q)^3. \quad (1.19)$$

(The proof of this is easily achieved (see §3) on opening the parentheses in the definition of G .) In the uninteresting case that $q > x$ the sum over j is empty and we immediately can finish the problem with, for instance, a corollary of Proposition 1.

COROLLARY. *We have, for $q > x$,*

$$G(x, q) = x \log x - x - \frac{x^2}{\phi(q)} + O(x(\log x)^{-A}) + O((\log q)^3).$$

In the case of interest, $q \leq x$, the formula (1.18) makes it clear that what is needed are good estimates for the twin prime sum $\sum_{N_1 < n \leq N_2} \Lambda(n) \Lambda(n + jq)$ at least on average over j . The assumption (1.17) more than suffices provided $q \geq x^{\frac{1}{2} + \epsilon}$. Lavrik [15] has given the proof of an average result of the type needed and Montgomery [16] used this in his work on $H(x, Q)$. We shall have need of a result stronger than Lavrik's that follows under GRH.

PROPOSITION 2. *Assume the Generalized Riemann Hypothesis. Then*

$$\sum_{0 < |k| \leq x} \left(\sum_{N_1(k) < n \leq N_2(x, k)} \Lambda(n) \Lambda(n + k) - \mathfrak{S}(k)(x - |k|) \right)^2 \ll x^{\frac{1}{2}} (\log x)^4. \quad (1.20)$$

Proposition 2 is the analogue of a theorem of Hardy and Littlewood (as sharpened by Goldston [8]) on the number of representations of even integers as the sum of two primes. We use (1.20) in place of (1.17) in Proposition 1 in our proof of Theorem 1. We also need to sum the average of the singular series that results once we apply (1.17) or (1.20). To do this we need the following result that slightly improves on corresponding statements of Montgomery [16] and Croft [3].

PROPOSITION 3. *We have*

$$\sum_{j \leq y} (y - j) \mathfrak{S}(jq) = \frac{y^2}{2} \frac{q}{\phi(q)} - \frac{y}{2} \log y - \frac{y}{2} \left(\gamma + \log 2\pi - 1 + \sum_{p|q} \frac{\log p}{p-1} \right) + I_\delta(y, q),$$

for any δ , $0 < \delta < \frac{1}{2}$, where, letting $2_q = \frac{2}{(2, q)}$,

$$I_\delta(y, q) = 2_q \mathfrak{S}(2_q q) \frac{1}{2\pi i} \int_{-\delta - i\infty}^{-\delta + i\infty} \zeta(s) \prod_{p \nmid 2q} \left(1 + \frac{1}{(p-2)p^s} \right) \left(\frac{y}{2_q} \right)^{s+1} \frac{ds}{s(s+1)},$$

and satisfies the bound

$$I_\delta(y, q) \ll \mathfrak{S}(2q)y^{1-\delta}\delta^{-1}\left(\frac{1}{2}-\delta\right)^{-\frac{1}{2}}\prod_{p|q}\left(1+\frac{1}{p^{1-\delta}}\right)\left(1+\frac{1}{p^{2(1-\delta)}}\right). \quad (1.21)$$

We may choose δ as we wish in $I_\delta(y, q)$, and this choice (see § 2) will depend on the range of y we are considering. We shall also, in § 6, sum $I_\delta(y, q)$ over q which smooths out occasional bad q .

We know of two different methods for proving asymptotic versions of the Barban–Davenport–Halberstam Theorem such as (1.7) and (1.8). The first of these, due to Montgomery [16], begins with Proposition 1 and uses Lavrik's result.

An elegant alternative approach, due to Hooley, also begins with Proposition 1. Now however, instead of summing over n and then q , one sums first over q . For fixed j and n the relevant sum is thus of the form $\sum_{m=n(j)} \Lambda(m)$. Since q is large, it follows that j is small and one may

appeal to the prime number theorem for arithmetic progressions in either its GRH or Siegel–Walfisz form.

As Hooley pointed out, both approaches give (1.7) but the former offers the advantage of allowing one to prove results for almost all moduli by considering the sum $\sum_q |E(x, jq)|$. Hooley [11] used his

approach to prove (1.8) and states [10, p 208] that the Montgomery method seems to require $Q > x^{\frac{1}{2}+\epsilon}$. While this certainly appears to be the case as long as one insists on estimating $\sum_q |E(x, jq)|$, the fact is that, for

the purpose of the Barban type sum, one requires only an estimate for $\sum_q E(x, jq)$ without absolute values and that allows us to give by the first method a new proof of (1.8), in slightly stronger form, which we believe to be simpler than that in [11]. This method for proving (1.8) is implicit in work of Ozlük [18] concerning the pair correlation of zeros of L -functions.

THEOREM 4. *Assuming the Generalized Riemann Hypothesis, we have, for $x^{\frac{1}{2}} \leq Q \leq x$,*

$$H(x, Q) = Qx \log Q - cQx + O(\min(Q^{\frac{1}{2}}x^{\frac{1}{2}}(\log x)^{\frac{1}{2}}, Qx)) + O(x^{\frac{1}{2}}(\log x)^6),$$

with

$$c = \gamma + \log 2\pi + 1 + \sum_p \frac{\log p}{p(p-1)}.$$

There are a number of closely related sums that could have been studied in place of $G(x, q)$, with essentially the same results, obtained by slight modifications to the definition of $E(x; q, a)$ given in (1.2). One obvious candidate is the replacement of $\psi(x; q, a)$ by $\theta(x; q, a)$ which

restricts the sum to primes rather than prime powers. Another change would be the replacement of the main term $\frac{x}{\phi(q)}$ in (1.2) by $\frac{\psi(x)}{\phi(q)}$ or, even better, $\frac{\psi(x, \chi_0)}{\phi(q)}$. These last changes lead to cleaner versions of Proposition 1 without the error term $\frac{2x}{\phi(q)} R(x)$, which at best is artificial and, when $q < x^{\frac{1}{2}}$, is a real impediment. They are however cosmetic; under GRH they alter G by less than the expected main term $x \log q$.

We are unable to give a good lower bound, even on average, when $q \leq x^{\frac{1}{2}}$. If q is very small, say $\log q = o(\log x)$, then we can at least say that (1.5) is not far from best possible, since by Cauchy's inequality and, say, E. Schmidt's omega theorem (cf [14, Theorem 33]),

$$\begin{aligned} G(x, q) &\geq \frac{1}{\phi(q)} \left(\sum_{a(q)}^* |E(x; q, a)| \right)^2 \geq \frac{1}{\phi(q)} |\psi(x) - x + O((\log x)^2)|^2 \\ &= \Omega\left(\frac{x}{\phi(q)}\right). \end{aligned} \quad (1.22)$$

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2. A singular series average

In this section we prove Proposition 3. We have

$$\begin{aligned} \sum_{j \leq y} (y-j) \mathfrak{S}(jq) &= 2C \sum_{\substack{j \leq y \\ jq \text{ even}}} (y-j) \prod_{\substack{p|jq \\ p>2}} \left(\frac{p-1}{p-2}\right) \\ &= 2C \sum_{\substack{j \leq y \\ jq \text{ even}}} (y-j) \prod_{\substack{p|q \\ p>2}} \left(\frac{p-1}{p-2}\right) \prod_{\substack{p|j \\ p \nmid 2q}} \left(\frac{p-1}{p-2}\right) \\ &= \begin{cases} \mathfrak{S}(q) \sum_{j \leq y} (y-j) \prod_{\substack{p|j \\ p \nmid 2q}} \left(\frac{p-1}{p-2}\right), & \text{if } q \text{ is even,} \\ \mathfrak{S}(2q) \sum_{j \leq \frac{y}{2}} 2\left(\frac{y}{2} - j\right) \prod_{\substack{p|j \\ p \nmid 2q}} \left(\frac{p-1}{p-2}\right), & \text{if } q \text{ is odd;} \end{cases} \\ &= 2_q \mathfrak{S}(2_q q) F\left(\frac{y}{2_q}, q\right), \end{aligned}$$

where

$$F(x, q) = \sum_{j \leq x} (x-j) \prod_{\substack{p|j \\ p \nmid 2q}} \left(\frac{p-1}{p-2}\right) = \sum_{j \leq x} (x-j) \sum_{d|j} f_q(d)$$

and

$$f_q(d) = \begin{cases} \mu^2(d) \prod_{p|d} (p-2)^{-1}, & \text{if } (d, 2q) = 1 \\ 0, & \text{if } (d, 2q) > 1. \end{cases}$$

Now, letting

$$F(s) = \prod_{p \nmid 2q} \left(1 + \frac{1}{(p-2)p^s}\right) = \sum_{d=1}^{\infty} \frac{f_q(d)}{d^s},$$

we have

$$F(x, q) = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} \zeta(s) F(s) \frac{x^{s+1}}{s(s+1)} ds.$$

Now

$$\begin{aligned} F(s) &= \prod_{p>2} \left(1 + \frac{1}{(p-2)p^s}\right) \prod_{\substack{p|q \\ p>2}} \left(1 + \frac{1}{(p-2)p^s}\right)^{-1} \\ &= \zeta(s+1) \left(1 - \frac{1}{2^{s+1}}\right) \prod_{p>2} \left(1 + \frac{1}{(p-2)p^s}\right) \\ &\quad \times \left(1 - \frac{1}{p^{s+1}}\right) \prod_{\substack{p|q \\ p>2}} \left(1 + \frac{1}{(p-2)p^s}\right)^{-1} \\ &= \zeta(s+1) G(s) H_q(s), \quad \text{say.} \end{aligned}$$

We see that $G(s)$ converges absolutely for $\sigma > -\frac{1}{2}$. On moving the contour to $s = -\delta + it$, $0 < \delta < \frac{1}{2}$, we encounter the poles at $s = 1$ and

$s = 0$ of $\zeta(s)\zeta(s+1)G(s)H_q(s) \frac{x^{s+1}}{s(s+1)}$. At $s = 1$, the residue is

$$= \frac{x^2}{2} \zeta(2) G(1) H_q(1) = \frac{x^2}{2} \prod_{p>2} \frac{(p-1)^2}{p(p-2)} \prod_{\substack{p|q \\ p>2}} \frac{p(p-2)}{(p-1)^2}.$$

The residue at $s = 0$ is

$$= x \zeta(0) G(0) H_q(0) \left(\gamma + \frac{\zeta'(0)}{\zeta(0)} + \frac{G'(0)}{G(0)} + \frac{H_q'(0)}{H_q(0)} + \log x - 1 \right).$$

In this we substitute the values

$$\zeta(0) = -\frac{1}{2}, \quad \frac{\zeta'}{\zeta}(0) = \log 2\pi, \quad G(0) = \frac{1}{2} \prod_{p>2} \frac{(p-1)^2}{p(p-2)}, \quad \frac{G'(0)}{G(0)} = \log 2,$$

$$H_q(0) = \prod_{\substack{p|q \\ p>2}} \left(\frac{p-2}{p-1} \right), \quad \frac{H_q'(0)}{H_q(0)} = \sum_{\substack{p|q \\ p>2}} \frac{\log p}{p-1},$$

and find that the residue at $s = 0$ is

$$= -\frac{x}{4} \left(\gamma + \log 2\pi + \log 2 + \sum_{\substack{p|q \\ p>2}} \frac{\log p}{p-1} + \log x - 1 \right) \prod_{p>2} \frac{(p-1)^2}{p(p-2)} \prod_{\substack{p|q \\ p>2}} \left(\frac{p-2}{p-1} \right).$$

When q is even the contribution to $\mathfrak{S}(q)F(y, q)$ from these two residues is

$$= y^2 \prod_{\substack{p|q \\ p>2}} \left(\frac{p}{p-1} \right) - \frac{y}{2} \left(\gamma + \log 2\pi + \log 2 + \sum_{\substack{p|q \\ p>2}} \frac{\log p}{p-1} + \log y - 1 \right).$$

Since q is even we have,

$$\prod_{\substack{p|q \\ p>2}} \left(\frac{p}{p-1} \right) = \frac{1}{2} \frac{q}{\phi(q)},$$

and also, the $\log 2$ may be included in the sum if the condition $p > 2$ is dropped. Thus the above contribution simplifies to

$$= \frac{y^2}{2} \frac{q}{\phi(q)} - \frac{y}{2} \log y - \frac{y}{2} \left(\gamma + \log 2\pi + \sum_{p|q} \frac{\log p}{p-1} - 1 \right).$$

When q is odd the contribution to $2\mathfrak{S}(2q)F(\frac{y}{2}, q)$ from the residues is

$$= \frac{y^2}{2} \prod_{\substack{p|q \\ p>2}} \left(\frac{p}{p-1} \right) - \frac{y}{2} \left(\gamma + \log 2\pi + \log 2 + \sum_{\substack{p|q \\ p>2}} \frac{\log p}{p-1} + \log \frac{y}{2} - 1 \right)$$

and since q is odd, this is also equal to the previous result.

We have thus obtained the first part of Proposition 3, and it remains to examine $I_\delta(y, q)$. For $0 < \delta < \frac{1}{2}$, we have

$$I_\delta(y, q)$$

$$\ll \mathfrak{S}(2q) \prod_{p|q} \left(1 + \frac{1}{p^{1-\delta}} \right) \left(1 + \frac{1}{p^{2(1-\delta)}} \right) y^{1-\delta} \int_0^\infty \left| \frac{\zeta(-\delta + it)\zeta(1-\delta + it)}{(-\delta + it)(1-\delta + it)} \right| dt$$

We first estimate the integral. For small t we have

$$\int_0^2 \ll \int_0^2 \frac{1}{\delta^2 + t^2} dt \ll \frac{1}{\delta}.$$

We know, for $0 < \delta < \frac{1}{2}$, that $|\zeta(-\delta + it)| \ll \delta^{-1} t^{\frac{1}{2}+\delta}$ and, by a theorem of Littlewood, cf [19, Theorem 7.2(A)],

$$\int_2^T |\zeta(1-\delta + it)|^2 dt \ll T \min \left(\log T, \left(\frac{1}{2} - \delta \right)^{-1} \right).$$

Hence by the Cauchy-Schwarz inequality, the remaining part of the integral in this case is

$$\begin{aligned} &<< \int_2^\infty \frac{|\zeta(1-\delta+it)|}{\delta t^{\frac{1}{2}-\delta}} dt << \delta^{-1} \left(\int_2^\infty \frac{1}{t^{\frac{1}{2}-\delta}} dt \int_2^\infty \frac{|\zeta(1-\delta+it)|^2}{t^{\frac{1}{2}-\delta}} dt \right)^{\frac{1}{2}} \\ &<< \delta^{-1} (\tfrac{1}{2} - \delta)^{-\frac{1}{2}}. \end{aligned}$$

We conclude that (1.21) holds, completing the proof of Proposition 3.

We shall need, for q large, upper bounds for the product over $p|q$ in (1.21). For the first factor in this product, we have

$$\begin{aligned} \prod_{p|q} \left(1 + \frac{1}{p^{1-\delta}} \right) &<< \exp \left(\sum_{p|q} \frac{1}{p^{1-\delta}} \right) << \exp \left(\sum_{p \leq 2 \log q} \frac{1}{p^{1-\delta}} \right) \\ &<< \exp \left(\int_2^{2 \log q} \frac{u^{\delta-1}}{\log u} du \right) << \exp \left(\int_{\delta \log 2}^{2\delta \log \log q} \frac{e^v}{v} dv \right). \end{aligned}$$

Now, if $2\delta \log \log q \geq 1$, the expression above is

$$\begin{aligned} &<< \exp \left(\int_{\delta \log 2}^1 \frac{1}{v} dv + \int_1^{2\delta \log \log q} \frac{e^v}{v} dv \right) \\ &<< \exp \left(\log \frac{1}{\delta} + \frac{e^{2\delta \log \log q}}{\delta \log \log q} \right) << \frac{1}{\delta} \exp \left(\frac{(\log q)^{2\delta}}{\delta \log \log q} \right), \end{aligned}$$

and in particular, if $\delta \log \log q \approx 1$ it is $<< \log \log q$.

For the second factor in the product we use the bounds

$$\prod_{p|q} \left(1 + \frac{1}{p^{2(1-\delta)}} \right) << \prod_p \left(1 + \frac{1}{p^{2(1-\delta)}} \right) << \sum_n \frac{1}{n^{2(1-\delta)}} << (\tfrac{1}{2} - \delta)^{-1}.$$

We conclude, using the bound $\mathfrak{S}(2q) << \log \log q$, that

$$I_\delta(y, q) << y(\log \log q)^3 \quad \text{if} \quad \delta = \frac{1}{\log \log q}, \quad (2.1)$$

that

$$I_\delta(y, q) << \mathfrak{S}(2q) \frac{y^{1-\delta}}{\delta^2} \exp \left(\frac{(\log q)^{2\delta}}{\delta \log \log q} \right) \quad \text{if} \quad \frac{1}{\log \log q} \leq \delta \leq \tfrac{1}{4}, \quad (2.2)$$

and

$$I_\delta(y, q) << y^{\frac{1}{2}} \exp \left(\frac{\log y}{\log q} \right) q^\epsilon \quad \text{if} \quad \delta = \tfrac{1}{2} - \frac{1}{\log q}. \quad (2.3)$$

3. Proof of Proposition 1

We have

$$G(x, q) = \sum_{a(q)}^* (\psi(x; q, a))^2 - \frac{2x}{\phi(q)} \sum_{a(q)}^* \psi(x; q, a) + \frac{x^2}{\phi(q)}.$$

Since

$$\begin{aligned} \sum_{a(q)}^* \psi(x; q, a) &= \sum_{\substack{n \leq x \\ (n, q)=1}} \Lambda(n) = \psi(x) + O\left(\log x \sum_{p|q} 1\right) \\ &= x + R(x) + O((\log x)^2), \end{aligned}$$

and

$$\sum_{a(q)}^* (\psi(x; q, a))^2 = \sum_{a(q)} (\psi(x; q, a))^2 + O((\log qx)^3),$$

we have

$$\begin{aligned} G(x, q) &= \sum_{a(q)} (\psi(x; q, a))^2 - \frac{x^2}{\phi(q)} - \frac{2x}{\phi(q)} R(x) \\ &\quad + O\left(\frac{x(\log x)^2}{\phi(q)}\right) + O((\log qx)^3). \end{aligned} \quad (3.1)$$

Now,

$$\sum_{a(q)} (\psi(x; q, a))^2 = \sum_{n \leq x} \Lambda^2(n) + \sum_{0 < |j| \leq \frac{x}{q}} \sum_{N_1(jq) < n \leq N_2(x, jq)} \Lambda(n) \Lambda(n + jq),$$

and Proposition 1 follows from this and the equation

$$\begin{aligned} \sum_{n \leq x} \Lambda^2(n) &= \sum_{n \leq x} \Lambda(n) \log n + O(x^{\frac{1}{2}} \log x) = \int_1^x \log u d\psi(u) + O(x^{\frac{1}{2}} \log x) \\ &= x \log x - x + O\left(\left(x^{\frac{1}{2}} + \max_{u \leq x} |R(u)|\right) \log x\right). \end{aligned}$$

We now apply Propositions 1 and 3 to immediately obtain the following result.

LEMMA 3.1. *We have for $1 \leq q \leq x$,*

$$\begin{aligned} G(x, q) &= x \log q - x \left(\gamma + \log 2\pi + \sum_{p|q} \frac{\log p}{p-1} \right) + \sum_{0 < |j| \leq \frac{x}{q}} E(x, jq) \\ &\quad - \frac{2x}{\phi(q)} R(x) + E + O\left(\min_{0 < \delta < \frac{1}{2}} q \left| I_\delta\left(\frac{x}{q}, q\right) \right| \right), \end{aligned}$$

with E satisfying (1.19).

Using the estimates (2.1) and (2.3) we see, for $x^\epsilon \leq q \leq x$ with any fixed $\epsilon > 0$,

$$\min_{0 < \delta < \frac{1}{2}} q I_\delta\left(\frac{x}{q}, q\right) \ll \min(x^{\frac{1}{2}} q^{\frac{1}{2} + \epsilon}, x(\log \log q)^3). \quad (3.2)$$

4. Proof of Theorem 3

We are assuming the estimates $E(x, k) \ll x^{\frac{1}{2}+\epsilon}$, $R(x) \ll x^{\frac{1}{2}+\epsilon}$; thus by Lemma 3.1 and (3.2),

$$G(x, q) = x \log q - x \left(\gamma + \log 2\pi + \sum_{p|q} \frac{\log p}{p-1} \right) + O(q^{-1} x^{\frac{1}{2}+\epsilon}) \\ + O(\min(x^{\frac{1}{2}} q^{\frac{1}{2}+\epsilon}, x(\log \log q)^3)). \quad (4.1)$$

Since

$$\sum_{p|q} \frac{\log p}{p-1} \ll \sum_{p < 2 \log q} \frac{\log p}{p} \ll \log \log q,$$

we see that, for $x^{\frac{1}{2}+\epsilon} \leq q \leq x$,

$$G(x, q) = x \log q + O(x(\log \log q)^3), \quad (4.2)$$

and Conjecture 2 follows. For $q = x^\alpha$, $\frac{1}{2} < \alpha < 1$, (4.1) gives

$$G(x, q) = x \log q - x \left(\gamma + \log 2\pi + \sum_{p|q} \frac{\log p}{p-1} \right) + O(x^{\max(\frac{1}{2}-\alpha, \frac{1}{2}+\frac{1}{2}\alpha)+\epsilon}), \quad (4.3)$$

yielding Conjecture 1.

5. Proof of Proposition 2

We write $N = [x]$, the integer part. Letting

$$S(\alpha) = \sum_{n \leq N} \Lambda(n) e(n\alpha), \quad (5.1)$$

then, with N_1, N_2 as in (1.15),

$$|S(\alpha)|^2 = \sum_{|k| \leq N} \left(\sum_{N_1(k) < n \leq N_2(N, k)} \Lambda(n) \Lambda(n+k) \right) e(k\alpha).$$

Letting $\beta = \alpha - \frac{b}{r}$,

$$I(\alpha) = \sum_{k \leq N} e(k\alpha), \quad J(\beta) = \frac{\mu(r)}{\phi(r)} I(\beta), \quad (5.2)$$

we have

$$|J(\alpha)|^2 = \sum_{|k| \leq N} \max(N - |k|, 0) e(k\alpha).$$

We also set

$$S_1(\beta; r, b) = S\left(\frac{b}{r} + \beta\right) - J(\beta), \quad (5.3)$$

and define

$$U(\alpha) = \sum_{0 < |k| \leq N} \mathfrak{S}(k) \max(N - |k|, 0) e(k\alpha).$$

We are now ready to prove Proposition 2. By (1.16) and Parseval's equation

$$\sum_{0 < |k| \leq N} (E(x, k))^2 = \int_0^1 \left| |S(\alpha)|^2 - U(\alpha) - \sum_{n \leq N} \Lambda^2(n) \right|^2 d\alpha.$$

Since $|a + b|^2 \leq 2(|a|^2 + |b|^2)$, the right hand side is

$$\leq \int_0^1 \left| |S(\alpha)|^2 - U(\alpha) \right|^2 d\alpha + O(N^2(\log N)^2).$$

We decompose the range of integration into Farey arcs $M_R(r, b) = \left(\frac{b+b'}{r+r'}, \frac{b+b''}{r+r''} \right]$ where $1 \leq r \leq R$, $(b, r) = 1$ and $\frac{b'}{r'} < \frac{b}{r} < \frac{b''}{r''}$ are neighbours in the decomposition. We let $\theta_R(r, b)$ be the translated interval $\left(\frac{-1}{r(r+r')}, \frac{1}{r(r+r'')} \right]$ and note that $\left(\frac{-1}{2rR}, \frac{1}{2rR} \right] \subset \theta_R(r, b) \subset \left(\frac{-1}{rR}, \frac{1}{rR} \right]$.

We write

$$\sum = \sum_R \sum_{r \leq R}^* \sum_{b(r)}^*, \quad \theta = \theta_R(r, b)$$

and have, with obvious additional abbreviations,

$$\begin{aligned} \int_0^1 \left| |S(\alpha)|^2 - U(\alpha) \right|^2 d\alpha &= \sum_R \int_{\theta} \left| |S|^2 - U \right|^2 = \sum_R \int_{\theta} \left| |S|^2 - |J|^2 + |J|^2 - U \right|^2 \\ &\leq 2 \sum_R \int_{\theta} \left| |S|^2 - |J|^2 \right|^2 + 2 \sum_R \int_{\theta} \left| |J|^2 - U \right|^2. \end{aligned} \quad (5.4)$$

To handle these terms we require the following lemma.

LEMMA 5.1(a) *We have*

$$\sum_R \int_{\theta} \left| |J|^2 - U \right|^2 \ll (1 + R^{-4} N^2) R^2 N \log N. \quad (5.5)$$

(b) *Assume GRH. Then*

$$\max_{\substack{r \leq R \\ (b, r) = 1}} \max_{\substack{|\beta| \leq \frac{1}{rR}}} |S_1(\beta; r, b)| \ll (1 + RN^{-\frac{1}{2}} \log N) R^{-\frac{1}{2}} N \log N. \quad (5.6)$$

Proof. Part (b) is Lemma 5 of [8]. Part (a) follows with tiny changes from the argument of [8, pp 150–151] which dealt with the analogous Goldbach sum.

From (5.6), the first sum on the right in (5.4) is seen to be

$$\begin{aligned} &\leq 2 \sum_R \int_{\theta} |S^2 - J^2|^2 = 2 \sum_R \int_{\theta} |S - J|^2 |S + J|^2 \\ &\ll \max_{\substack{r \leq R \\ (b, r) = 1}} \max_{|\beta| \leq \frac{1}{2h}} |S_1(\beta; r, b)|^2 \int_0^1 (|S|^2 + |J|^2) \\ &\ll N^{\frac{1}{2}} (\log N)^4, \end{aligned}$$

on choosing $R = N^{\frac{1}{2}} (\log N)^{-1}$. By (5.5), with this choice of R we have

$$\sum_R \int_{\theta} ||J|^2 - U|^2 \ll N^2 (\log N)^3.$$

This proves the proposition. .

6. Proof of Theorem 1

We first need the following result on $I_{\delta}(x, q)$ on average over q .

LEMMA 6.1. *We have, for $\frac{1}{2} < \delta < 1$, and $Q \leq x$*

$$\sum_{Q \leq q \leq 2Q} q \left| I_{\delta} \left(\frac{x}{q}, q \right) \right| \ll Q^{1+\delta} x^{1-\delta} \delta^{-1} \left(\frac{1}{2} - \delta \right)^{-\frac{1}{2}}. \quad (6.1)$$

Also, for any $\epsilon > 0$ and $x^{\epsilon} \leq Q \leq x$,

$$\min_{0 < \delta < \frac{1}{2}} \sum_{Q \leq q \leq 2Q} q \left| I_{\delta} \left(\frac{x}{q}, q \right) \right| \ll \min \{ Q^{\frac{1}{2}} x^{\frac{1}{2}} (\log Q)^{\frac{1}{2}}, QX \}. \quad (6.2)$$

Proof. We use the fact (see [16] or [6]) that $\sum_{q \leq Q} \mathfrak{S}(q) \sim Q$. By (1.21) we see the left hand side of (6.1) is

$$\begin{aligned} &\ll \sum_{Q \leq q \leq 2Q} q \mathfrak{S}(2q) \left(\frac{x}{q} \right)^{1-\delta} \delta^{-1} \left(\frac{1}{2} - \delta \right)^{\frac{1}{2}} \prod_{p|q} \left(1 + \frac{1}{p^{1-\delta}} \right) \left(1 + \frac{1}{p^{2(1-\delta)}} \right) \\ &\ll Q^{\delta} x^{1-\delta} \delta^{-1} \left(\frac{1}{2} - \delta \right)^{-\frac{1}{2}} \sum_{q \leq 2Q} \mathfrak{S}(2q) \prod_{p|q} \left(1 + \frac{1}{p^{1-\delta}} \right) \left(1 + \frac{1}{p^{2(1-\delta)}} \right). \end{aligned}$$

Equation (6.1) now follows since the sum above is

$$\begin{aligned} &\ll \sum_{q \leq 2Q} \mathfrak{S}(2q) \sum_{d|q} \frac{1}{d^{1-\delta}} \leq \sum_{d \leq 2Q} \frac{\mathfrak{S}(2d)}{d^{1-\delta}} \sum_{r \leq \frac{2Q}{d}} \mathfrak{S}(2r) \\ &\ll Q \sum_{d \leq 2Q} \frac{\mathfrak{S}(2d)}{d^{2-\delta}} \ll Q. \end{aligned}$$

To prove (6.2), note $Q^{1+\delta}x^{1-\delta} = Qx(Q/x)^\delta$. Hence we do best if we chose δ as large as possible unless Q is very close to x . The choices $\delta = \frac{1}{2} - \frac{1}{\log Q}$ (since $Q \geq x^\epsilon$) and $\delta = \frac{3}{4}$ in (6.1) give (6.2).

We now prove Theorem 1. First assume GRH. By Lemma 3.1 the left hand side of (1.11) is

$$\ll \sum_{\frac{x}{Q} < q \leq Q} \sum_{0 < |j| \leq \frac{x}{q}} |E(x, jq)| + \min_{0 < \delta \leq \frac{1}{2}} \sum_{\frac{x}{Q} < q \leq Q} q \left| I_\delta\left(\frac{x}{q}, q\right) \right| + x^{\frac{3}{2}}(\log x)^3. \quad (6.3)$$

By Proposition 2 and Cauchy's inequality, the first error is

$$\begin{aligned} &\leq \sum_{q \leq x} \sum_{0 < |jq| \leq x} |E(x, jq)| \leq \sum_{0 < |k| \leq x} \tau(|k|) |E(x, k)| \\ &\ll \left(\sum_{k \leq x} \tau^2(k) \sum_{0 < |k| \leq x} |E(x, k)|^2 \right)^{\frac{1}{2}} \ll x^{\frac{7}{2}}(\log x)^{\frac{7}{2}}. \end{aligned} \quad (6.4)$$

We use (6.2) for the second error in (6.3) and the proof is complete. We have actually proved (1.11) for $q \geq x^\epsilon$. However, by (1.5) the left side of (1.11) is $\ll Qx(\log x)^4$ which is better than (1.11) for $Q \leq x^{\frac{1}{2}}(\log x)^{-\frac{1}{2}}$.

To prove the unconditional version of Theorem 1 we merely use a weak form of the prime number theorem which gives (6.3) with the last term replaced by $x^2(\log x)^{-A}$ and then apply Lavrik's result, see [16, Lemma 1], to get this weaker bound also in (6.4).

7. Proof of Theorem 4

In this section we give our proof of Hooley's GRH version of the Barban–Montgomery asymptotic formula. Our work in this section was motivated by [18]. The main result we need is the following estimate.

PROPOSITION 4. *For $Q \geq x^{\frac{1}{2}}$, we have on GRH,*

$$\sum_{0 < |j| \leq \frac{x}{Q}} \left| \sum_{Q < q \leq \frac{x}{|j|}} E(x, jq) \right| \ll x^{\frac{1}{2}}(\log x)^6, \quad (7.1)$$

where $E(x, jq)$ is given by (1.16).

We remark that, for application to Theorem 4 we do not require the

absolute value signs in (7.1) but they come to us at no cost. Using Proposition 1 it can be shown on GRH (and probably without it) that (7.1) is best possible apart from the logarithmic factor.

Proof of Theorem 4. By (1.5) we have, assuming GRH, that

$$\begin{aligned} H(x, Q) &= \sum_{q \leq Q} G(x, q) = \sum_{q \leq x^{\frac{1}{2}}} G(x, q) + \sum_{x^{\frac{1}{2}} < q \leq Q} G(x, q) \\ &= \sum_{x^{\frac{1}{2}} < q \leq Q} G(x, q) + O(x^{\frac{1}{2}}(\log x)^4). \end{aligned} \quad (7.2)$$

Hence, by Lemma 3.1 and RH,

$$\begin{aligned} H(x, Q) &= \sum_{x^{\frac{1}{2}} < q \leq Q} \left(x \log q - x \left(\gamma + \log 2\pi + \sum_{p|q} \frac{\log p}{p-1} \right) \right) \\ &\quad + \sum_{x^{\frac{1}{2}} < q \leq Q} \sum_{0 < |j| \leq \frac{x}{q}} E(x, jq) \\ &\quad + O \left(\sum_{x^{\frac{1}{2}} < q \leq Q} \min_{0 < \delta < \frac{1}{2}} q \left| I_{\delta} \left(\frac{x}{q}, q \right) \right| \right) + O(x^{\frac{1}{2}}(\log x)^4). \end{aligned}$$

In the first sum we may extend the summation range to $1 \leq q < Q$ with an error of $O(x^{\frac{1}{2}} \log x)$, and then the resulting sum is

$$\begin{aligned} &= x \log ([Q]!) - x[Q](\gamma + \log 2\pi) - x \sum_{q \leq Q} \sum_{p|q} \frac{\log p}{p-1} \\ &= Qx \log Q - Qx(\gamma + \log 2\pi + 1) + O(x \log Q) - x \sum_{p \leq Q} \frac{\log p}{p-1} \left[\frac{Q}{p} \right] \\ &= Qx \log Q - Qx \left(\gamma + \log 2\pi + 1 + \sum_p \frac{\log p}{p(p-1)} \right) + O(x \log Q). \end{aligned}$$

Next, the sum involving E is

$$\begin{aligned} &= \sum_{x^{\frac{1}{2}} < q \leq x} \sum_{0 < |j| \leq \frac{x}{q}} E(x, jq) - \sum_{Q < q \leq x} \sum_{0 < |j| \leq \frac{x}{q}} E(x, jq) \\ &\ll \sum_{0 < |j| \leq x^{\frac{1}{2}}} \left| \sum_{x^{\frac{1}{2}} < q \leq \frac{x}{|j|}} E(x, jq) \right| + \sum_{0 < |j| \leq \frac{x}{Q}} \left| \sum_{Q < q \leq \frac{x}{|j|}} E(x, jq) \right| \\ &\ll x^{\frac{1}{2}}(\log x)^6, \end{aligned}$$

where we have used Proposition 4 in the last line twice, with $Q = x^{\frac{1}{2}}$ and with Q itself. Finally, applying Lemma 6.1 to the error term involving $I_{\delta} \left(\frac{x}{2}, q \right)$ over ranges $\left(\frac{Q}{2^{n+1}}, \frac{Q}{2^n} \right]$ and summing over n we find this term is

$$\ll \min(Q^{\frac{1}{2}} x^{\frac{1}{2}} (\log Q)^{\frac{1}{2}}, Qx).$$

This proves Theorem 4.

To prove Proposition 4, we begin with some lemmata.

LEMMA 7.1. For $\delta \geq x^{-1}$, S_1 defined by (5.3), we have, on GRH,

$$\sum_{b(r)}^* \int_{-\delta}^{\delta} |S_1(\beta; r, b)|^2 d\beta \ll \delta r x (\log rx)^4. \quad (7.3)$$

Proof. This is proven, for instance, as formula (7.15) in [7] except that there the sum S runs only over primes. The contribution here coming from high powers of primes is $\ll \delta r x$ which is admissible.

LEMMA 7.2. Let $\left| \alpha - \frac{b}{r} \right| \leq \frac{1}{r^2}$, $(b, r) = 1$. Then, for $r \leq x^{\frac{1}{2}} \leq Q$, we have

$$\sum_{j < \frac{x}{Q}} \left| \sum_{Q < q \leq x} e(jq\alpha) \right| \ll r^{-1} x \log x. \quad (7.4)$$

Proof. This is well known, essentially due to Vinogradov and Vaughan. An argument is given for example in [4, pp. 143–144].

Proof of Proposition 4. Since $E(x, -k) = E(x, k)$ it suffices to restrict the outer summation to positive j . We start with the identity

$$\sum_{n \leq x - jq} \Lambda(n) \Lambda(n + jq) = \int_0^1 |S(\alpha)|^2 e(jq\alpha) d\alpha, \quad (7.5)$$

and take a Farey decomposition of order R . The integral in (7.5) is thus

$$\sum_{r \leq R} e\left(\frac{jqb}{r}\right) \sum_{b(r)}^* \int_{\theta_R(r, b)} \left| S\left(\frac{b}{r} + \beta\right) \right|^2 e(jq\beta) d\beta. \quad (7.6)$$

We write $S = J + (S - J)$ in the above equation and find,

$$\sum_{n \leq x - jq} \Lambda(n) \Lambda(n + jq) = I_1 + 2I_2 + I_3 \quad (7.7)$$

where

$$I_1 = \sum_{r \leq R} \sum_{b(r)}^* e\left(\frac{jqb}{r}\right) \int_{\theta} |J|^2 e(jq\beta) d\beta, \quad (7.8)$$

$$I_2 = \sum_{r \leq R} \sum_{b(r)}^* e\left(\frac{jqb}{r}\right) \int_{\theta} \operatorname{Re}(J(\overline{S - J})) e(jq\beta) d\beta, \quad (7.9)$$

$$I_3 = \sum_{r \leq R} \sum_{b(r)}^* e\left(\frac{jqb}{r}\right) \int_{\theta} |S - J|^2 e(jq\beta) d\beta. \quad (7.10)$$

Estimation of I_3 : Writing the left hand side of (7.4) as $W(\alpha)$ we have

$$\sum_{j < \frac{x}{Q}} \left| \sum_{Q < q \leq \frac{x}{j}} I_3 \right| = \sum_{r \leq R} \sum_{b(r)}^* \int_{\theta} |S - J|^2 W\left(\frac{b}{r} + \beta\right) d\beta.$$

We choose $R \leq x^{\frac{1}{2}}$ so that by Lemma 7.2 this is

$$\ll x \log x \sum_{r \leq R} \frac{1}{r} \sum_{b(r)}^* \int_{-\frac{1}{rR}}^{\frac{1}{rR}} |S - J|^2 d\beta,$$

and, by Lemma 7.1 we deduce that, on GRH,

$$\sum_{j < \frac{x}{Q}} \left| \sum_{Q < q \leq \frac{x}{j}} I_3 \right| \ll R^{-1} x^2 (\log x)^6. \quad (7.11)$$

Estimation of I_2 : We again apply Lemma 7.2 to deduce that

$$\sum_{j < \frac{x}{Q}} \left| \sum_{Q < q \leq \frac{x}{j}} I_2 \right| \ll x \log x \sum_{r \leq R} \frac{\mu^2(r)}{r \phi(r)} \sum_{b(r)}^* \int_{\theta} |I| |S - J|. \quad (7.12)$$

Here I , given in (5.2), satisfies $I \ll \min(x, \|\beta\|^{-1})$ and so by the Cauchy-Schwarz inequality

$$\sum_{b(r)}^* \int_{-\frac{1}{x}}^{\frac{1}{x}} |I| |S - J| \ll x \left(\sum_{b(r)}^* \int_{-\frac{1}{x}}^{\frac{1}{x}} |S - J|^2 \right)^{\frac{1}{2}} \left(\frac{r}{x} \right)^{\frac{1}{2}} \ll r x^{\frac{1}{2}} (\log x)^2,$$

by Lemma 7.1. This gives a contribution to (7.12) which is $\ll x^{\frac{3}{2}} (\log x)^4$.

Next we note that, for $u_j = (2^j r R)^{-1}$, we have

$$\begin{aligned} \sum_{b(r)}^* \int_{\frac{1}{2}u_j}^{u_j} |I| |S - J| &\ll u_j^{-1} \sum_{b(r)}^* \int_{-u_j}^{u_j} |S - J| \\ &\ll u_j^{-1} (u_j r)^{\frac{1}{2}} \left(\sum_{b(r)}^* \int_{-u_j}^{u_j} |S - J|^2 \right)^{\frac{1}{2}} \\ &\ll r x^{\frac{1}{2}} (\log x)^2 \end{aligned}$$

by Lemma 7.1. Thus, this term also contributes $\ll x^{\frac{3}{2}} (\log x)^4$ to (7.12) and since there are $\ll \log x$ such terms we conclude

$$\sum_{j < \frac{x}{Q}} \left| \sum_{Q < q \leq \frac{x}{j}} I_2 \right| \ll x^{\frac{3}{2}} (\log x)^5. \quad (7.13)$$

Estimation of I_1 : Write

$$\int_{\theta_R(r,b)} = \int_0^1 - \int_{[0,1] \setminus \theta_R(r,b)} \quad (7.14)$$

We have, with $c_r(n) = \sum_{d|r, d|n} d\mu(d)$ the Ramanujan sum, and other obvious notation,

$$\begin{aligned} \sum_{r \leq R} \sum_{b(r)}^* e\left(\frac{jqb}{r}\right) \int_0^1 |J|^2 e(jq\beta) d\beta &= \sum_{r \leq R} \frac{\mu^2(r)}{\phi^2(r)} c_r(jq) \sum_{\substack{n, m \leq x \\ n-m=jq}} 1 \\ &= \mathfrak{S}_R(jq) \sum_{m \leq x-jq} 1 = (\mathfrak{S}(jq) - \mathfrak{S}_R(jq))(x-jq) + O(|\mathfrak{S}_R(jq)|). \end{aligned} \quad (7.15)$$

From (7.14) and (7.15) we deduce that

$$I_1 = \mathfrak{S}(jq)(x-jq) + O(\log jq) + E_1 + E_2 \quad (7.16)$$

where

$$E_1 = - \sum_{r \leq R} \sum_{b(r)}^* e\left(\frac{jqb}{r}\right) \int_{[0,1] \setminus \theta_R(r,b)} |J|^2 e(jq\beta) d\beta$$

and

$$E_2 = -\mathfrak{S}_R(jq)(x-jq).$$

We now show that

$$\sum_{j \leq \frac{x}{Q}} \left| \sum_{Q < q \leq \frac{x}{j}} E_2 \right| \ll R^{-1} x^2 (\log x)^3. \quad (7.17)$$

The left side is

$$\begin{aligned} &\ll x \sum_{jq \leq x} \sum_{r > R} \frac{\mu^2(r)}{\phi^2(r)} \sum_{\substack{d|r \\ d|jq}} d \ll x \sum_{\substack{jq \leq x \\ d|jq}} \frac{d\mu^2(d)}{\phi^2(d)} \sum_{\substack{r_1 > \frac{x}{d} \\ d|r_1}} \frac{\mu^2(r_1)}{\phi^2(r_1)} \\ &\ll \frac{x}{R} \sum_{\substack{jq \leq x \\ d|jq}} \frac{d^2 \mu^2(d)}{\phi^2(d)} = \frac{x}{R} \sum_{d \leq x} \frac{d^2 \mu^2(d)}{\phi^2(d)} \sum_{\substack{m \leq x \\ m \equiv 0(d)}} \tau(m) \\ &\leq \frac{x}{R} \sum_{d \leq x} \frac{d^2 \mu^2(d) \tau(d)}{\phi^2(d)} \sum_{j \leq \frac{x}{d}} \tau(j) \ll R^{-1} x^2 (\log x)^3. \end{aligned}$$

Next we show

$$\sum_{j \leq \frac{x}{Q}} \left| \sum_{Q < q \leq \frac{x}{j}} E_1 \right| \ll Rx \log x. \quad (7.18)$$

The left hand side is

$$\ll \sum_{r \leq R} \sum_{b(r)}^* \frac{\mu^2(r)}{\phi^2(r)} \int_{[0,1] \setminus \theta_R(r,b)} |I(\beta)|^2 \left| W\left(\frac{b}{r} + \beta\right) \right| d\beta. \quad (7.19)$$

We now take a Farey decomposition of $[0,1]$ of order S where $x^{\frac{1}{2}} \geq S > R$. Then, comparing the arcs around $\frac{b}{r}$, we have $\theta_R(r,b) \supset \theta_S(r,b)$, and so

$$\begin{aligned} & \int_{[0,1] \setminus \theta_R(r,b)} |I(\beta)|^2 \left| W\left(\frac{b}{r} + \beta\right) \right| d\beta \\ &= \int_{[0,1] \setminus \mathcal{M}_R(r,b)} \left| I\left(\alpha - \frac{b}{r}\right) \right|^2 |W(\alpha)| d\alpha \leq \int_{[0,1] \setminus \mathcal{M}_S(r,b)} \left| I\left(\alpha - \frac{b}{r}\right) \right|^2 |W(\alpha)| d\alpha \\ &= \sum_{s \leq S} \sum_{\substack{c(s) \\ \frac{c}{s} \neq \frac{b}{r}}}^* \int_{\mathcal{M}_S(s,c)} \left| I\left(\alpha - \frac{b}{r}\right) \right|^2 |W(\alpha)| d\alpha \\ &= \sum_{s \leq S} \sum_{\substack{c(s) \\ \frac{c}{s} \neq \frac{b}{r}}}^* \int_{\theta_S(s,c)} \left| I\left(\frac{c}{s} - \frac{b}{r} + \beta\right) \right|^2 \left| W\left(\frac{c}{s} + \beta\right) \right| d\beta \\ &\ll x \log x \sum_{s \leq S} \frac{1}{s} \sum_{\substack{c(s) \\ \frac{c}{s} \neq \frac{b}{r}}}^* \int_{\theta_S(s,c)} \left| I\left(\frac{c}{s} - \frac{b}{r} + \beta\right) \right|^2 d\beta. \end{aligned} \quad (7.20)$$

Now, on $\theta_S(s,c)$,

$$\left| I\left(\frac{c}{s} - \frac{b}{r} + \beta\right) \right| \ll \left\| \frac{c}{s} - \frac{b}{r} + \beta \right\|^{-1} \ll \left\| \frac{cr - bs + rs\beta}{rs} \right\|^{-1}$$

and

$$rs\beta \leq \frac{rs}{sS} \leq \frac{r}{S} \leq \frac{1}{2} \quad \text{if } S \geq 2R.$$

Thus

$$\left\| \frac{cr - bs + rs\beta}{rs} \right\|^{-1} \leq 2 \left\| \frac{cr - bs}{rs} \right\|^{-1}$$

and the integral (7.20) is bounded as

$$\ll x \log x \sum_{r \leq S} \frac{1}{s} \sum_{c(s)}^* \frac{1}{sS} \frac{(rs)^2}{|cr - bs|^2},$$

where, here and in the next few lines, we adopt the convention that for

each pair of residue classes $b(r), c(s)$, representative integers are chosen so as to minimize $|cr - bs|$. Summing over b , then r , we majorize (7.19) by

$$\ll \frac{x \log x}{S} \sum_{r \leq R} \frac{\mu^2(r)r^2}{\phi^2(r)} \sum_{s \leq S} \left(\sum_{\substack{b(r) \\ \frac{b}{r} \neq \frac{s}{r}}}^* \sum_{\substack{c(s) \\ \frac{c}{s} \neq \frac{b}{r}}}^* \frac{1}{|cr - bs|^2} \right). \quad (7.21)$$

Here for given r, s , the number of representations (by residue classes b, c) of an integer m in the form $m = cr - bs$ is just (r, s) if m is divisible by (r, s) and is zero otherwise. Thus we have

$$\sum_{\substack{b(r) \\ \frac{b}{r} \neq \frac{s}{r}}}^* \sum_{\substack{c(s) \\ \frac{c}{s} \neq \frac{b}{r}}}^* \frac{1}{|cr - bs|^2} \leq \frac{2}{(r, s)} \zeta(2)$$

which, with (7.21), gives (7.18). By (7.5), (7.7), and (7.16) we have shown that

$$E(x, jq) = 2I_2 + I_3 + E_1 + E_2 + O(\log jq).$$

Combining (7.13), (7.11), (7.18), (7.17), and choosing $S = 2R = x^{\frac{1}{2}}$, we complete the proof of Proposition 4 and hence of Theorem 4.

We remark that it is implicit in the proof that the error term in Theorem 4 can be improved slightly in narrow ranges near $Q = x^{\frac{1}{2}}$ and near $Q = x$.

8. Proof of Theorem 2

Our work in this section is motivated by [9]. Let

$$\psi_R(x; q, a) = \sum_{\substack{n \leq x \\ n \equiv a(q)}} \lambda_R(n),$$

where

$$\lambda_R(n) = \sum_{r \leq R} \frac{\mu^2(r)}{\phi(r)} c_r(n), \quad (8.1)$$

and let

$$L(R) = \sum_{r \leq R} \frac{\mu^2(r)}{\phi(r)} = \log R + O(1).$$

We require the following lemma of [9].

LEMMA 8.1. For $1 \leq R \leq x$, we have

$$\sum_{n \leq x} \lambda_R(n) \Lambda(n) = \psi(x) L(R) + O(R \log x), \quad (8.2)$$

$$\sum_{n \leq x} \lambda_R^2(n) = x L(R) + O(R^2), \quad (8.3)$$

and for $0 < |k| \leq x$, with N_1, N_2 as in (1.15), and E given in (1.2),

$$\begin{aligned} \sum_{N_1(k) < n \leq N_2(x, k)} \lambda_R(n) \Lambda(n+k) &= \mathfrak{S}(k)(x - |k|) + O\left(\frac{k\tau(k)x}{\phi(k)R}\right) \\ &+ O\left(\sum_{r \leq R} \frac{\mu^2(r)r \log(2R/r)}{\phi(r)} |E(N_2 + k; r, k) - E(N_1 + k; r, k)|\right), \end{aligned} \quad (8.4)$$

and

$$\sum_{N_1(k) < n \leq N_2(x, k)} \lambda_R(n) \lambda_R(n+k) = \mathfrak{S}(k)(x - |k|) + O\left(\frac{k\tau(k)x}{\phi(k)R}\right) + O(R^2). \quad (8.5)$$

The lower bound of Theorem 2 is based on the inequality

$$\begin{aligned} 0 &\leq \sum_{a=1}^q (\psi(x; q, a) - \psi_R(x; q, a))^2 \\ &= \sum_{a=1}^q (\psi(x; q, a))^2 - 2 \sum_{a=1}^q \psi(x; q, a) \psi_R(x; q, a) + \sum_{a=1}^q (\psi_R(x; q, a))^2 \\ &= \sum_{a=1}^q (\psi(x; q, a))^2 - \sum_{\substack{n, m \leq x \\ n \equiv m(q)}} (2\Lambda(n)\lambda_R(m) - \lambda_R(n)\lambda_R(m)). \end{aligned}$$

Hence,

$$\begin{aligned} \sum_{a=1}^q (\psi(x; q, a))^2 &\geq \sum_{\substack{n, m \leq x \\ n \equiv m(q)}} (2\Lambda(n) - \lambda_R(n)) \lambda_R(m) \\ &= \sum_{n \leq x} (2\Lambda(n) - \lambda_R(n)) \lambda_R(n) \\ &\quad + \sum_{0 < |j| \leq \frac{x}{q}} \sum_{N_1(jq) < n \leq N_2(x, jq)} (2\Lambda(n) - \lambda_R(n)) \lambda_R(n + jq). \end{aligned}$$

By Lemma 8.1 we have, since $\mathfrak{S}(-n) = \mathfrak{S}(n)$,

$$\sum_{a=1}^q (\psi(x; q, a))^2 \geq x \log R + 2 \sum_{j \leq \frac{x}{q}} (x - jq) \mathfrak{S}(jq) + E,$$

where

$$E \ll R(x) \log R + R \log x + x + \frac{x}{q} R^2 + \frac{x}{R} \sum_{j \leq \frac{x}{q}} \frac{jq \tau(jq)}{\phi(jq)} \\ + \sum_{j \leq \frac{x}{q}} \sum_{r \leq R} \frac{\mu^2(r) r \log(2R/r)}{\phi(r)} |E(N_2 + jq; r, jq) - E(N_1 + jq; r, jq)|. \quad (8.6)$$

In the unconditional case we are taking $x(\log x)^{-A} \leq q \leq x$, and so, by the Bombieri-Vinogradov theorem we may take $R = x^{\frac{1}{2}}(\log x)^{-B}$ with B sufficiently large that the last double sum in (8.6) is $\ll q^{-1}x^2(\log x)^{-A} \ll x$ and then, since we have $R(x) \ll x(\log x)^{-1}$,

$$E \ll x + \frac{\tau(q) x^2 \log x}{\phi(q) R} \ll x + q^{-1+\epsilon} x^{\frac{3}{2}+\epsilon} \ll x.$$

On GRH the estimate (1.3) implies the last double sum in (8.6) is $\ll q^{-1} R x^{\frac{3}{2}} (\log x)^2$, and on choosing $R = qx^{-\frac{1}{2}} (\log x)^{-2}$ we find that

$$E \ll x + \frac{q}{(\log x)^4} + \frac{\tau(q) x^{\frac{3}{2}} (\log x)^3}{\phi(q) q}$$

which, for $x^{\frac{3}{2}+\epsilon} \leq q \leq x$, gives $E \ll x$. We conclude that

$$\sum_{a=1}^q (\psi(x; q, a))^2 \geq (\tfrac{1}{2} - \epsilon) x \log x + 2 \sum_{j \leq \frac{x}{q}} (x - jq) \mathfrak{S}(jq) \quad \text{for } x(\log x)^{-A} \leq q \leq x,$$

and on GRH,

$$\sum_{a=1}^q (\psi(x; q, a))^2 \geq x \log \left(\frac{q}{x^{\frac{1}{2}} (\log x)^2} \right) + O(x) + 2 \sum_{j \leq \frac{x}{q}} (x - jq) \mathfrak{S}(jq) \quad \text{for } x^{\frac{3}{2}+\epsilon} \leq q \leq x.$$

Thus by (3.1),

$$G(x, q) \geq (\tfrac{1}{2} - \epsilon) x \log x + 2q \sum_{j \leq \frac{x}{q}} \left(\frac{x}{q} - j \right) \mathfrak{S}(jq) - \frac{x^2}{\phi(q)} \quad \text{for } x(\log x)^{-A} \leq q \leq x,$$

and on GRH,

$$G(x, q) \geq (\alpha - \tfrac{1}{2} - \epsilon) x \log x + 2q \sum_{j \leq \frac{x}{q}} \left(\frac{x}{q} - j \right) \mathfrak{S}(jq) - \frac{x^2}{\phi(q)} \quad \text{for } x^{\frac{3}{2}+\epsilon} \leq q \leq x.$$

An application of Proposition 3 and (2.1) now gives Theorem 2.

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